MIND MAPS

for

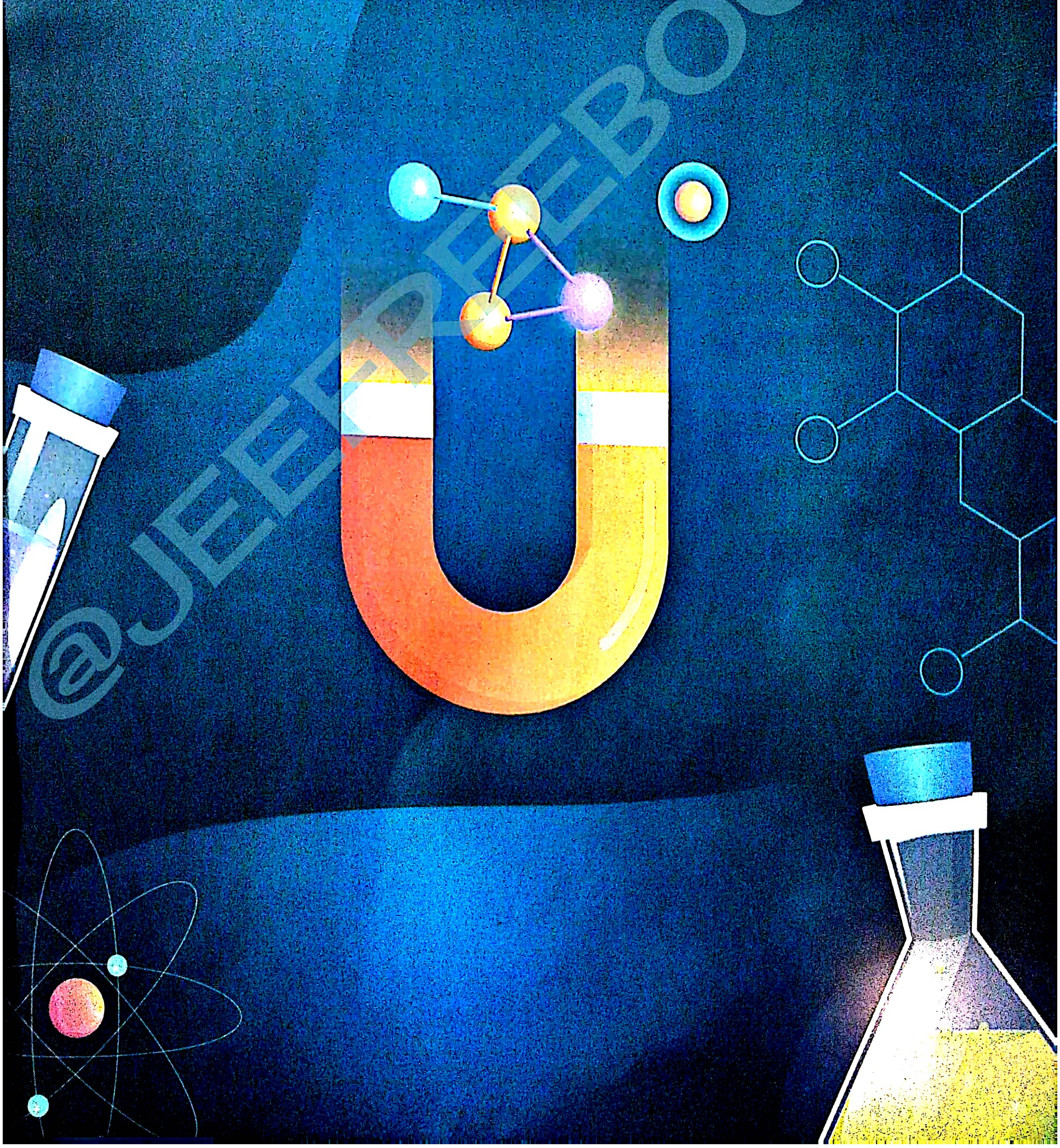
JEE

CLASS-12TH

CONTENTS

MIND MAP

PHYSICS



ELECTROSTATICS - 1

Charge

Quantization of charge

$Q = \pm ne$ Q=Total charge

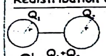
n=1,2,3....

$e = 1.6 \times 10^{-19} C$

Additivity of charge

$Q' = Q_1 + Q_2$

Redistribution of charge



$Q' = \frac{Q_1 + Q_2}{2}$

Q'=Charge on each shell after redistribution

Charge Density

Linear Charge density, $\lambda = \frac{Q}{L}$ Unit= $\frac{C}{m}$

Surface Charge density, $\sigma = \frac{Q}{S}$ Unit= $\frac{C}{m^2}$

Volume Charge density, $\rho = \frac{Q}{V}$ Unit= $\frac{C}{m^3}$

Q=Total charge V=Volume

L=Length S=Area

?

If a charge on the body is 1 nC, then how many electrons are present on the body?

a) 1.6×10^{19} b) 6.25×10^{9}

c) 6.25×10^{27} d) 6.25×10^{28}

Coulomb's Law

$Q_1 \quad Q_2 \quad F = \frac{1}{4\pi\varepsilon_0} \frac{Q_1 Q_2}{r^2}$

ε_0 = Permitivity of free space

$[\varepsilon_0] = \frac{[Q_1][Q_2]}{[r^2][F]} = \frac{[AT][AT]}{[L^2][MLT^{-2}]} = M^{-1}L^{-3}T^4A^2$

$F_{med} = \frac{F_{air}}{k}$

$Q_1 \xrightarrow{F} k \xleftarrow{F} Q_2$

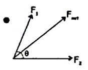
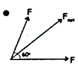
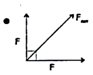
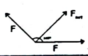
k=dielectric constant of the medium

Superposition

Direction:

a) Like- Towards the point at which force has to be evaluated (repulsion)

b) Unlike- Away from the point at which force has to be evaluated (attraction)

-  General rule $F_{Net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$
-  When $\theta = 60°$ $F_{net} = \sqrt{3}F$
-  When $\theta = 90°$ $F_{net} = \sqrt{2}F$
-  When $\theta = 120°$ $F_{net} = F$

Equilibrium of Charges

Calculation of Charge

$Q_1 \overset{r_1}{—} q \overset{r_2}{—} Q_2$

$\frac{Q_1}{Q_2} = \left(\frac{r_1}{r_2}\right)^2$ q in equilibrium

$q = -\left(\frac{r_1}{r_1 + r_2}\right)^2 Q_2$ Q_1 in equilibrium

$q = -\left(\frac{r_2}{r_1 + r_2}\right)^2 Q_1$ Q_2 in equilibrium

?

A charge is placed at the centre of the line joining two equal charges Q. The system of the three charges will be in equilibrium if q is equal to

a) -Q/2 c) +Q/4

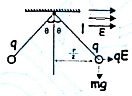
b) -Q/4 d) +Q/2

Charge on pendulum

$\tan\theta = \frac{qE}{mg}$

$\sin\theta = \frac{r}{2l}$

$r = 2l\sin\theta$

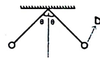


if θ is very small

$\tan\theta \simeq \sin\theta$

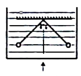
$\frac{r}{2l} = \frac{qE}{mg}$

$\frac{r}{2l} = \frac{kq^2/r^2}{mg}$, $r^3 \propto q^2$



Density of ball "ρ"

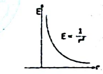
it θ does not change on submerging in liquid Dielectric constant of liquid,



density of liquid=σ

$K = \frac{\rho}{\rho - \sigma}$

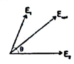
Electric Field



Electric field at a point, due to point charge $E = \frac{Kq}{r^2}$

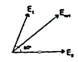
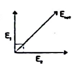
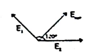
$K = \frac{1}{4\pi\varepsilon_0}$

Superposition



General rule

$E_{Net} = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos\theta}$

- $E_{Net} = E_1 + E_2$ (same direction)
- $E_{Net} = |E_1 - E_2|$ (opposite direction)
-  $E_{Net} = \sqrt{E_1^2 + E_2^2 + E_1E_2}$ If, $E_1 = E_2 = E$ Then, $E_{net} = \sqrt{3}E$
-  $E_{Net} = \sqrt{E_1^2 + E_2^2}$ If, $E_1 = E_2 = E$ Then, $E_{net} = \sqrt{2}E$
-  $E_{Net} = \sqrt{E_1^2 + E_2^2 - E_1E_2}$ If, $E_1 = E_2 = E$ Then, $E_{Net} = E$

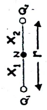
Direction

1) Positive charge:-Towards the point at which electric field has to be evaluated

2) Negative charge:-Away from the point at which electric field has to be evaluated

Neutral Point

Like Charges



$x_1 = \frac{\sqrt{Q_1}\, r}{\sqrt{Q_1} + \sqrt{Q_2}}$

$x_2 = \frac{\sqrt{Q_2}\, r}{\sqrt{Q_1} + \sqrt{Q_2}}$

Unlike Charges

Outside closer to smaller charge

$|Q_2| < |Q_1|$

$x = \frac{\sqrt{Q_2}\, r}{\sqrt{Q_1} - \sqrt{Q_2}}$

Distance from $Q_1 = x + r$

?

Two point charges +8q and -2q are located at x = 0 and x = L respectively. The location of a point on the x axis at which the net electric field due to these two point charges is zero is:

a) 8L

b) 4L

c) 2L

d) L/4

Charged particle released in an electric field

1) Force, $F = qE$

2) Acceleration, $a = \frac{qE}{m}$

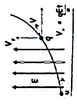
3) Velocity, $V = \frac{qE}{m}t$

4) Velocity, $V = \sqrt{\frac{2qE}{m}x}$

5) Kinetic energy, $K.E = \frac{q^2E^2t^2}{2m}$

accelerated in the direction of electric field

accelerated opposite to the direction of electric field



$V = \sqrt{V_x^2 + V_y^2}$

$= \sqrt{U^2 + \left(\frac{qE}{m}t\right)^2}$

accelerated in the direction of field and perpendicular to initial velocity

$\frac{M_p}{M_e} = 1837$, $\frac{e}{m} = 1.7 \times 10^{-11}$

$\frac{1}{2}at^2 = h = \text{Constant}$

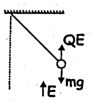
$\frac{1}{2}\frac{qE}{m}t^2 = h$

$t^2 \propto m$

$\frac{t_p}{t_e} = \left|\frac{m_p}{m_e}\right|^{1/2}$

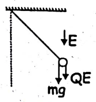
$\Rightarrow t_p > t_e$

Time period of Charged Pendulum in an electric filed



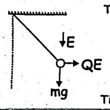
$T=2\pi\sqrt{\frac{l}{(g-\frac{QE}{m})}}$

Time period will increase



$T=2\pi\sqrt{\frac{l}{(g+\frac{QE}{m})}}$

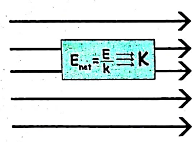
Time period will decrease



$T=2\pi\sqrt{\frac{l}{\sqrt{(g^2+(\frac{QE}{m})^2)}}}$

Time period will decrease

Electric field inside a dielectric medium

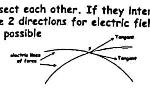


$E_{net}=E-E_{induced}$, $E_{induced}=E-E_{net}=E(1-\frac{1}{K})$

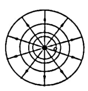
Properties of field lines

1) Start from positive charge and end on negative charge

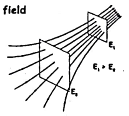
2) Never intersect each other. If they intersect there will be 2 directions for electric field which is not possible



3) Always perpendicular to Conducting surface

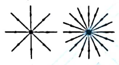


4) $E \propto$ Electric field line density



5) Never form closed loops (Conservative force)

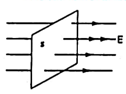
6) $q \propto$ no. of field lines $|q_2|>|q_1|$



? Electric lines of force about negative point charge are:

a) circular, anticlockwise
b) circular, clockwise
c) radial, inward
d) radial, outward

Electric flux



Flux is proportional to total no. of field lines passing through an area

$\Phi=\int E.ds\cos\theta$

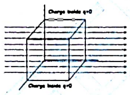
$\Phi=\int \vec{E}.\vec{ds}$

Gauss Law:- $\Phi=\frac{q}{\varepsilon_0}=\oint E.ds\cos\theta$

Zero flux:- $\Phi=\frac{q_{net}}{\varepsilon_0}=0$, where $q_{net}=0$

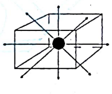
Electric flux for Cube

1) No charge inside the cube

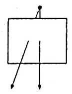


$\Phi=\frac{q}{\varepsilon_0}=0$

2) Charge placed at the center

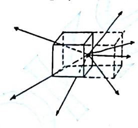


$\Phi_{total}=\frac{q}{\varepsilon_0}$



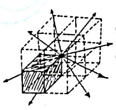
$\Phi_{one\ side}=\frac{q}{6\varepsilon_0}$

3) Charge placed at the face



$\Phi_{cube}=\frac{q}{2\varepsilon_0}$

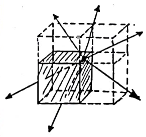
4) Charge placed at the corner



$\Phi_{cube}=\frac{Q}{8\varepsilon_0}$

$\Phi_{one\ face}=\frac{Q}{8\varepsilon_0}\times\frac{1}{3}=\frac{Q}{24\varepsilon_0}$

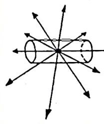
5) Charge placed at the edge



$\Phi=\frac{Q}{4}$

$\Phi_{face}=\frac{Q}{4\varepsilon_0}\times\frac{1}{4}=\frac{Q}{16\varepsilon_0}$

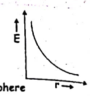
6) Flux through curved surface



$\Phi_{effective}=\Phi_{curve}+2\Phi_{cross\ section}$

Application of Gauss's Theorem

1) Point charge $E=\frac{Kq}{r^2}$

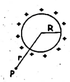


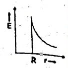
2) Metal sphere/Hollow sphere

$E_{surface}=\frac{KQ}{R^2}$

$E_{outside}=\frac{KQ}{r^2}$

$E_{inside}=0$



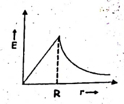


3) Non-Conducting sphere

$E_{inside}=\frac{KQr}{R^3}$

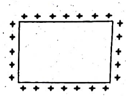
$E_{surface}=\frac{KQ}{R^2}$

$E_{outside}=\frac{KQ}{r^2}$



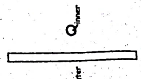
4) Conducting sheet

$E=\frac{\sigma}{\varepsilon_0}$



5) Non-conducting sheet

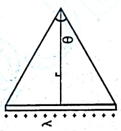
$E=\frac{\sigma}{2\varepsilon_0}$



Q_{outer} Q_{inner} Q_{plate}

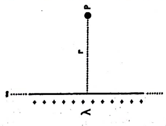
$Q_{inner}=Q_{plate}-Q_{outer}$

7) Electric field due to a finite linear charge distribution



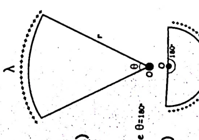
$E=\frac{2k\lambda}{r}\sin(\theta/2)$

8) Electric field due to a infinite linear charge distribution



$E=\frac{2k\lambda}{r}$

9) Electric field due to circular arc at its center



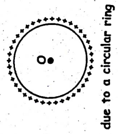
$E_o=\frac{2k\lambda}{r}\sin(\theta/2)$

eg: For a semicircle $\theta=180$

$E_o=\frac{2k\lambda}{r}\sin(180/2)$

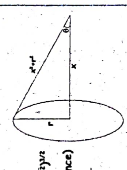
$=\frac{2k\lambda}{r}$

10) Electric field at the center of a circular ring



$E_o=0$

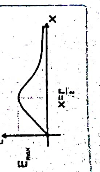
11) Electric field due to a circular ring of charge



$E=\frac{1}{4\pi\varepsilon_0}\frac{qx}{(x^2+r^2)^{3/2}}$

(For large distance)

$E=\frac{1}{4\pi\varepsilon_0}\frac{q}{x^2}$



ELECTROSTATICS-2

ELECTROSTATIC POTENTIAL ENERGY

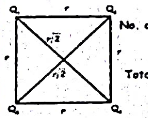
2 point charges

Q_1 —— r —— Q_2

$\Delta U = \frac{1}{4\pi\epsilon_0}\frac{Q_1 Q_2}{r}$ < like charges - positive (repulsive energy); Unlike charges - negative (attractive energy)

System of charges

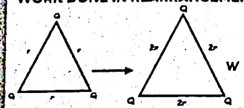
$\Delta U_{system} = \Sigma \Delta U_{pair}$ No. of pairs $= \frac{n(n-1)}{2}$



No. of pairs $= \frac{4 \times 3}{2} = 6$

Total P.E $= \frac{KQ_1Q_2}{r} + \frac{KQ_1Q_4}{r/2} + \frac{KQ_4Q_3}{r/2} + \frac{KQ_1Q_3}{r} + \frac{KQ_4Q_2}{r} + \frac{KQ_1Q_3}{r}$

WORK DONE IN REARRANGEMENT OF THE SYSTEM



$U_i = \frac{3kQ^2}{r}$ $U_f = \frac{3kQ^2}{2r}$

$W = U_f - U_i = \frac{3kQ^2}{2r} - \frac{3kQ^2}{r} = -\frac{3kQ^2}{2r}$

ELECTROSTATIC POTENTIAL

$V_p = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$

$V_{AB} = V_B - V_A = \frac{Q}{4\pi\epsilon_0}\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$

WORK DONE

$w = q[V_B - V_A]$

$w = 1 \times (-500-500) = -1000J$

Superposition of potential - Algebraic sum of all potentials

ZERO POTENTIAL

a) Like charge-no zero potential point

b) Unlike charge- 2 points of zero potential on line joining

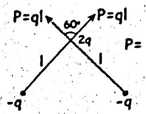
Inside point $= a = \frac{Q_1 r}{Q_1+Q_2}$ Outside point $= b = \frac{Q_1 r}{Q_1 - Q_2}$

DIPOLE

Dipole moment

$\vec{P} = q\, 2\vec{l}$

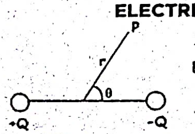
Direction is from -Q to +Q



$P = 2ql\cos30^\circ = 2\sqrt{3}ql(\uparrow)$

Q. Fine net dipole moment

ELECTRIC FIELD

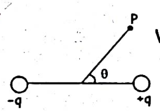


$E_p = \frac{Kp}{r^3}\sqrt{3\cos^2\theta + 1}$

$E_{axial} = \frac{Kp}{r^3}\sqrt{3+1} = \frac{2Kp}{r^3}$ 0° with dipole moment, E_{net} from -Q to +Q

$E_{equitorial} = \frac{Kp}{r^3}\sqrt{0+1} = \frac{Kp}{r^3}$ 180° with dipole moment, E_{net} from +Q to -Q

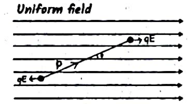
ELECTRIC POTENTIAL



$V_p = \frac{kp\cos\theta}{r^2}$

$V_{axial} = \frac{Kp}{r^2}$ $V_{equitorial} = 0$

TORQUE

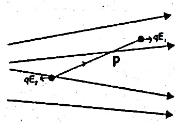


Uniform field

- Rotational motion only
- No translatory motion

$\vec{\tau} = \vec{p} \times \vec{E}$

$\tau = PE\sin\theta$



Non - Uniform field

- Both rotatory and translatory motion

$\tau_{max} = pE\sin\theta = pE, (\theta = 90^\circ)$

$\tau_{min} = pE\sin\theta = 0, (\theta = 0^\circ)$

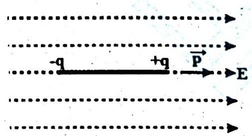
WORK DONE

$W = pE(\cos\theta_1 - \cos\theta_2)$

POTENTIAL ENERGY

$U = -pE\cos\theta$

1) $U_{minimum}$ (stable equilibrium)



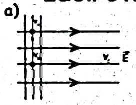
$\theta = 0^\circ$

$\cos\theta = 1$

$U = -pE$

EQUIPOTENTIAL SURFACE

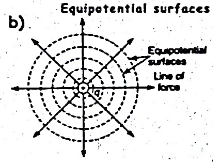
a)



$V_A = V_B > V_C$

Equipotential surfaces

b)



c) Field lines and equipotential surface are perpendicular to each other at every location.

d) Work done in moving a charge on equipotential surface is 0

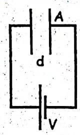
ELECTRIC FIELD & POTENTIAL

$E = \frac{-dV}{dr}$ $E_x = \frac{-\partial V}{\partial x}$ $E_y = \frac{-\partial V}{\partial y}$ $E_z = \frac{-\partial V}{\partial z}$

$\Delta V = -\int \vec{E}\cdot d\vec{r}$

CAPACITANCE

CAPACITANCE



$Q = CV$

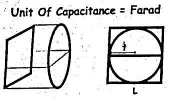
$C = \frac{Q}{V}$

$C = \frac{\varepsilon_0 A}{d}$

Capacitance Depends on
1. Distance betweent plates
2. Area of plates
3. Medium b/w plates

Capacitance is Independent of
1. Charge
2. Potential difference

Unit Of Capacitance = Farad



$A_{eff} = \pi\left(\frac{L}{2}\right)^2 = \pi\frac{L^2}{4}$

$C = \frac{\varepsilon_0 A_{eff}}{d} = \frac{\pi\varepsilon_0 L^2}{4d}$

ENERGY STORED

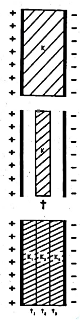
Work done by battery $= QV$ (100%)

- 50% Energy stored in capacitor $= \frac{1}{2}QV = \frac{CV^2}{2} = \frac{Q^2}{2C}$
- 50% Energy dissipated $= \frac{1}{2}QV = \frac{CV^2}{2} = \frac{Q^2}{2C}$

Variation with plate seperation

1. $Q = CV = \frac{\varepsilon_0 AV}{d}$
2. $C = \frac{\varepsilon_0 A}{d}$ $\quad C \propto \frac{1}{d}$
3. $V = \frac{Q}{C} = \frac{Qd}{\varepsilon_0 A}$ $\quad$ 4. $E = \frac{Q}{A\varepsilon_0} = \frac{V}{d}$
5. $F = \frac{Q^2}{2A\varepsilon_0} = \frac{CV^2}{2d}$
6. $U = \frac{CV^2}{2}$ OR $U = \frac{Q^2}{2C} = \frac{Q^2 d}{2\varepsilon_0 A} = \frac{\varepsilon_0 AV^2}{2d}$

DIELECTRIC IN CAPACITOR



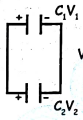
$C' = \frac{k\varepsilon_0 A}{d}$

$C' = KC$

$C' = \frac{\varepsilon_0 A}{d - t + \frac{t}{K}}$

$C' = \frac{\varepsilon_0 A}{\frac{t_1}{K_1} + \frac{t_2}{K_2} + \frac{t_3}{K_3}}$

Connecting two charged capacitors - Case 1



$V_{common} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

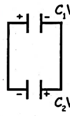
Initial Energy → Heat loss → Final Energy

$U_i = \frac{C_1V_1^2}{2} + \frac{C_2V_2^2}{2}$

$U_f = \frac{1}{2}(C_1 + C_2)\left(\frac{C_1V_1 + C_2V_2}{C_1 + C_2}\right)^2$

Energy loss $= \frac{C_1C_2}{2(C_1 + C_2)}(V_1 - V_2)^2$

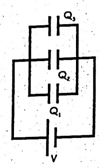
Connecting two charged capacitors - Case 2



$V_{common} = \frac{C_1V_1 - C_2V_2}{C_1 + C_2}$

Energy loss $= \frac{C_1C_2}{2(C_1 + C_2)}(V_1 + V_2)^2$

2. Parallel combination



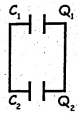
$C_{eq} = C_1 + C_2 + C_3$

$C_{eq} = nC$

[If $C_1 = C_2 = \ldots\ldots = C_n = C$]

Charge divider rule

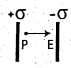
$Q \propto C$



$Q_1 = \left(\frac{C_1}{C_1 + C_2}\right)Q$

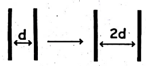
$Q_2 = \left(\frac{C_2}{C_1 + C_2}\right)Q$

ELECTRIC FIELD



$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0} = \frac{CV}{A\varepsilon_0}$

when plate seperation doubled

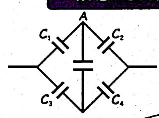


$C = \frac{\varepsilon_0 A}{d}$ $\quad C' = \frac{C}{2}$

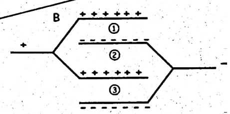
Dielectric inserted in capacitor

Battery removed	Battery remains connected
1. $C = KC$	1. $C' = KC$
2. $Q' = Q$	2. $Q' = KQ$
3. $V' = \frac{V}{K}$	3. $V' = V$
4. $E' = \frac{E}{K}$	4. $E' = E$
5. $U' = \frac{U}{K}$	5. $U' = KU$

WHEATSTONE'S BRIDGE



$\frac{C_1}{C_2} = \frac{C_3}{C_4}$ $\quad V_A = V_B$

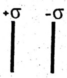


$C = \frac{3\varepsilon_0 A}{d}$

POTENTIAL

$V = Ed = \frac{Qd}{A\varepsilon_0}$

Force b/w plates of a parallel plate capacitor



$F = Q \times E_{\text{Due to one plate}}$

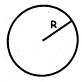
$= Q \times \frac{\sigma}{2\varepsilon_0} = Q \times \frac{Q}{2A\varepsilon_0}$

$= \frac{Q^2}{2A\varepsilon_0} = \frac{C^2V^2}{2A\varepsilon_0} = \frac{\varepsilon_0 A}{d} \times \frac{CV^2}{2A\varepsilon_0}$

$= \frac{CV^2}{2d}$

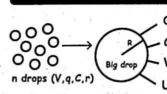
Battery connected $d \to 2d$	Battery Disconnected $d \to 2d$
1. V=Constant	1. Q=Constant
2. $C' = \frac{C}{2}$	2. $C' = \frac{C}{2}$
3. $Q = CV$, $Q' = \frac{Q}{2}$	3. $V = \frac{Q}{C}$, $V' = 2V$
4. $E = \frac{V}{d}$, $E' = \frac{E}{2}$	4. $E = \frac{Q}{A\varepsilon_0}$ = Constant
5. $F = \frac{CV^2}{2d}$, $F' = \frac{F}{4}$	5. $F = \frac{Q^2}{2A\varepsilon_0}$ = Constant
6. $U = \frac{CV^2}{2}$, $U' = \frac{U}{2}$	6. $U = \frac{Q^2}{2C}$, $U' = 2U$

SPHERICAL CAPACITOR



$C = 4\pi\varepsilon_0 R$

REDISTRIBUTION OF CHARGE



$Q' = nq$

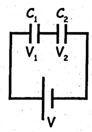
$C' = n^{1/3}C$

$V' = n^{2/3}V$

$U' = n^{5/3}U$

Grouping of capacitors

1. Series combination



$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$

$V = V_1 + V_2$

$C_{eq} = \left(\frac{C_1C_2}{C_1 + C_2}\right)$

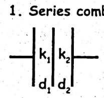
Voltage divider rule

$V \propto \frac{1}{C}$

$V_1 = \left(\frac{C_2}{C_1 + C_2}\right)V$ $\quad V_2 = \left(\frac{C_1}{C_1 + C_2}\right)V$

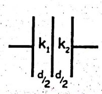
MULTIPLE DIELECTRICS

1. Series combination



$k_{eq} = \frac{\Sigma d}{\Sigma d/k}$

$k_{eq} = \frac{d_1 + d_2}{\frac{d_1}{k_1} + \frac{d_2}{k_2}}$

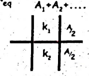


$k_{eq} = \frac{2k_1k_2}{k_1 + k_2}$

2. Parallel combination

$k_{eq} = \frac{\Sigma kA}{\Sigma A}$

$k_{eq} = \frac{k_1A_1 + k_2A_2 + \ldots}{A_1 + A_2 + \ldots}$



$k_{eq} = \frac{k_1\frac{A}{2} + k_2\frac{A}{2}}{\frac{A}{2} + \frac{A}{2}} = \frac{k_1 + k_2}{2}$

CURRENT ELECTRICITY

PHYSICS WALLAH

01 ELECTRIC CURRENT

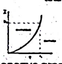
Current $I = \frac{q}{t}$

Average current $I_{av} = \frac{\Delta q}{\Delta t}$

Instantaneous current $I_{inst} = \frac{dq}{dt}$

Average current I_{av}

$= \frac{\text{area under I-t graph}}{\text{total time taken}}$

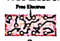


Concave area $= \frac{2}{3} x_0 y_0$

Convex area $= \frac{1}{3} x_0 y_0$

02 DRIFT VELOCITY

Free Electron



$\langle KE \rangle = \frac{3}{2} kT$

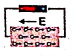
$\langle \frac{1}{2} mv^2 \rangle \approx 10^{-21} J$

Avg. Speed = 105 m/s

Electrons are in random motion

Avg. velocity $= \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \ldots \vec{v}_n}{n} = 0$

$I_{net} = 0$



E accelerates the electrons

$\vec{v} = \vec{u} + \vec{a}t$

$v_d = a\tau$

$v_d = \frac{eE}{m}\tau$

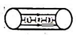
$E = \frac{V}{l}$

$v_d = \frac{eV}{ml}\tau$

03 FACTORS AFFECTING DRIFT VELOCITY

- Dependence on shape

1) Uniform shape



$v_d = \frac{eE}{m}\tau$

Here E is uniform so,

$v_{d1} = v_{d2} = v_{d3}$ $E_1 = E_2 = E_3$

2) Relation B/w Current & Drift velocity
$I = nAv_d e$

n= no. of e's per unit volume

04 OHM'S LAW

$V = I \times R$

Voltage volts (V) — Current Amperes (A) — Resistance ohms(Ω)

$T_1 > T_2$, Slope = $\tan\theta = R$

$R = \frac{\rho l}{A}$ — Depends on
1. Material (n & τ changes)
2. Temperature (n & τ changes)
3. Dimension (Length & Area)

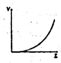
$\rho = \frac{RA}{l} = \frac{m}{ne^2\tau}$ — Depends on
1. Material (n & τ changes)
2. Temperature (n & τ changes)

Non-Ohmic Conductor

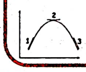
V-I graph is not linear

Slope of tangent $\frac{dV}{dI} = R$

Resistance is not constant



1) Slope=+ve
Resistance = +ve
V↑ then I↑

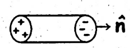


2) Slope=0
Resistance=0

3) Slope=-ve
Resistance = -ve
V↑ then I↓

05 CURRENT DENSITY

$J = \frac{I}{A} n$



$I = \frac{V}{R} = \frac{V}{} \frac{EA}{\rho}$

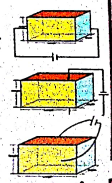
$J = \frac{E}{\rho}$, $J \propto E$

06 DEPENDANCE OF R ON DIMENSION

$R = \frac{\rho l}{A} = \frac{\rho l}{bh}$

$R = \frac{\rho h}{lb}$

$R = \frac{\rho b}{lh}$



$\frac{R_{max}}{R_{min}} = \frac{(\text{max length})^2}{(\text{min length})^2}$

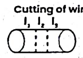
07 CUTTING & STRETCHING OF WIRE

$R_1 \propto l_1$

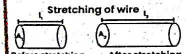
$R_2 \propto l_2$

$R_3 \propto l_3$

Cutting of wire



Stretching of wire



Before stretching After stretching

$\frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2 = \left(\frac{A_2}{A_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^4 = \left(\frac{d_2}{d_1}\right)^4$

If l become nl then R become $n^2 R$

If r become r/n then R become $n^4 R$

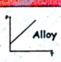
If change in length>10% $\frac{R_2 - R_1}{R_1} \times 100 = \frac{l_2^2 - l_1^2}{l_1^2} \times 100$

If change in length <10%

1) % change in R=2×% change in length
2) % change in R=2×% change in area
3) % change in R=4×% change in radius

08 TEMPERATURE DEPENDANCE OF RESISTANCE

Metals, α =+Ve
If T↑ R↑



For semiconductor α = -ve
If T↑, R↓

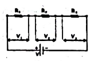
α= slightly increasing with temp.
Resistance slightly increases with T

Variation of resistance with temperature $\alpha = \frac{\Delta R}{R \Delta T}$

09 GROUPING OF RESISTANCE

Series Combination

Current is constant
voltage is divided
$R_s = R_1 + R_2 + R_3 + \ldots R_n$

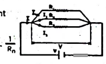


If resistors are identical: $R_s = nR$

Parallel Combination

voltage is constant
current is divided

$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_n}$



If resistors are identical: $R_p = \frac{R}{n}$

Shortcut for two resistors in parallel

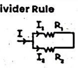
$\frac{R_1 R_2}{R_1 + R_2}$

$\left.\begin{matrix} R_1 \\ R_2 \end{matrix}\right\}$ ↗ R_s Bigger than largest value of resistance ↘ R_p Lower than smallest value of resistance

10 CURRENT & VOLTAGE DIVIDER RULE

Current Divider Rule

V =Constant $I \propto \frac{1}{R}$



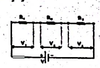
$I_1 = \frac{I \times R_2}{R_1 + R_2}$, $I_2 = \frac{I \times R_1}{R_1 + R_2}$

Voltage Divider Rule

$V_1 = IR_1$ $V_2 = IR_2$ $V_3 = I_3 R_3$

$V_1 = \frac{V \times R_1}{R_1 + R_2 + R_3}$

$V_2 = \frac{V \times R_2}{R_1 + R_2 + R_3}$

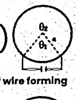


$V_3 = \frac{V \times R_3}{R_1 + R_2 + R_3}$

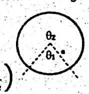
11 GEOMETRICAL DIAGRAM

Circle formed by wire having uniform resistance per unit length (r)

$R_{eff} = ra \left(\frac{\theta_1 \theta_2}{\theta_1 + \theta_2}\right)$



When resistance of wire forming circle is given

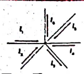


$R_{eff} = \frac{R}{2\pi} \left(\frac{\theta_1 \theta_2}{\theta_1 + \theta_2}\right)$

PHYSICS WALLAH

12 KIRCHHOFF'S LAW

1. Junction Rule

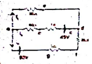


$$\sum I_{in} = \sum I_{out}$$
$$I_1 + I_2 + I_4 + I_5 = I_3 + I_6$$

2. Open Circuit

(1) A —B $V_A - V_B = -E$
(2) A —B $V_A - V_B = +E$
(3) A —B $V_A - V_B = IR$
(4) A —B $V_A - V_B = -IR$

Closed Circuit



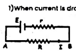
(1) abcdea
$40I_3 + I_1 - 45 + 30I_1 = 0$
(1) aedfga
$-30I_1 + 20I_2 + I_2 + 80 = 0$

13 CELL & INTERNAL RESISTANCE

$V_A - V_B = E - Ir$

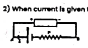
Terminal potential difference (TPD)

1) When current is drawn from cell



$V = V_A - V_B = E - Ir$
TPD< EMF
$V = E - Ir = IR \Rightarrow I = \frac{E}{R+r}$
So, $V = \frac{E}{R+r}R$

2) When current is given to cell



$V = V_A - V_B = E + Ir$
TPD> EMF

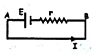
14 CELL & INTERNAL RESISTANCE

3) When cell is in open circuit

A —B Here, I=0

TPD=E

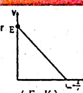
3) When cell is in short circuit



$I = I_{max} = \frac{E}{r}$
$TPD = 0$

15 CELL & INTERNAL RESISTANCE

Slope of graph =-r
y intercept=E

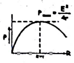


Internal Resistance $r = \left(\frac{E-V}{V}\right)R$

Power delivered by cell during withdrawl of current

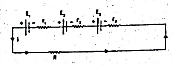
$P = I^2R = \left(\frac{E}{R+r}\right)^2 R$ $\quad P_{max} = \frac{E^2}{4r}$, when R=r

Maximum power transfered



16 COMBINATION OF CELLS

1) Series Combination

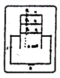


(a) $E_{equivalent} = E_1 + E_2 + E_3 + \_\_ + E_n$
(b) $r_{equivalent} = r_1 + r_2 + r_3 + \_\_ + r_n$
(c) Current, $I = \frac{\sum E_i}{\sum r_i + R}$
(d) If all cells have equal emf E and equal internal resistance r then $I = \frac{nE}{nr + R}$
1) If $nr >> R \Rightarrow I = \frac{E}{r}$
2) If $nr << R \Rightarrow I = \frac{nE}{R}$
(e) Power dissipated in circuit $P = I^2R = \left(\frac{nE}{nr+R}\right)^2 R$

Conditions for maximum power: R=nr
$P_{max} = nE^2/4r$

17 COMBINATION OF CELLS

2) Parallel Combination



a) $E_{equivalent} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3} \ldots \frac{E_n}{r_n}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \cdots \frac{1}{r_n}}$

b) $r_{equivalent} = \frac{1}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \cdots \frac{1}{r_n}}$

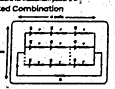
(c) If all cells have equal emf E & internal resistance r then $E_{equivalent} = E$

$r_{equivalent} = \frac{r}{n} \Rightarrow$ current $I = \frac{E}{\frac{r}{n} + R}$

(d) Power dissipated in the circuit $P = I^2R = \left(\frac{nE}{r+nR}\right)^2 R$

Conditions for maximum power $R = \frac{r}{n}$

3) Mixed Combination



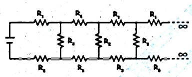
$r_{eq} = \frac{nr}{m}$ Total emf $= E_{eq} = nE$
$I = \frac{E_{eq}}{R + r_{eq}} = \frac{nE}{R + \frac{nr}{m}}$

$P_{max} = \frac{E_{eq}}{4R} = \frac{mnE^2}{4r}$ $\left[\because R = \frac{nr}{m}\right]$

n cells connected in series and there are m such branches in the circuit.
Internal resistance of cells connected in a row nr

18 COMBINATION OF CELLS

Infinite resistors

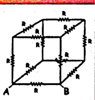


$$R_{eq} = \frac{R_1 + R_3}{2}\left[1 + \sqrt{1 + \frac{4R_2}{R_1 + R_3}}\right]$$

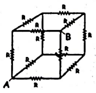
If all resistors are equal $R_{eq} = R(1+\sqrt{3})$

19 3D CIRCUIT

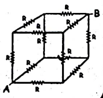
Edge $R_{eq} = \frac{7R}{12}$



Face $R_{eq} = \frac{3R}{4}$



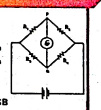
Body $R_{eq} = \frac{5R}{6}$



20 WHEATSTONE BRIDGE

1) Balanced WSB

Balanced Condition $\frac{R_1}{R_2} = \frac{R_3}{R_4}$
$V_P = V_Q$
Current through G = 0

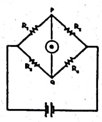


2) Unbalanced WSB

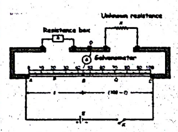
$V_P \neq V_Q$

if $\frac{R_1}{R_2} > \frac{R_3}{R_4}$

then $V_Q > V_P$



21 METER BRIDGE



$$\frac{P}{Q} = \frac{R}{X} = \frac{\ell}{100-\ell}$$